

The Modal Triplet Theory Program D1: A Projection-First Dark-Sector Hypothesis and Its Covariant Completion Contract

Peter Nero

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Abstract

This paper formulates a projection-first dark-sector hypothesis within Modal Triplet Theory (MTT) and identifies exactly what would be required to turn it into a physical model. The retained idea is that a reduced description may have an availability profile in which statistical or relational encodings remain usable while more local encodings do not. We prove, however, that an availability profile alone cannot determine a stress-energy tensor, a modified field equation, an equation of state, or a perturbation spectrum. In particular, statistical encoding does not imply cold, pressureless, collisionless matter, and relational encoding does not imply accelerated expansion.

We therefore replace the former identification of dark matter with E9 and dark energy with E8 by two explicit hypotheses. Each requires a selected covariant source map or a selected diffeomorphism-invariant effective action. We derive the associated conservation condition and state separate background, linear-perturbation, lensing, cluster-collision, cosmic-microwave-background, baryon-acoustic-oscillation, structure-growth, and equation-of-state gates. The current MTT result is consequently an organized candidate taxonomy and a completion contract, not a derivation of the observed dark sector. This revision records which earlier claims are retained, withdrawn, or left open.

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1 Status, Purpose, and Logical Tier

Observations constrain a successful cosmological model through several independent channels. The cosmic microwave background (CMB), baryon acoustic oscillations (BAO), weak lensing, galaxy clustering, supernova distances, and cluster collisions do not merely ask whether an additional gravitational effect exists; together they constrain its background evolution, perturbations, clustering, lensing response, and interactions [5, 1, 2, 3].

The purpose of this paper is narrower. It asks whether restrictions on the availability of reduced encodings could organize candidate explanations of dark-sector phenomenology. The answer is:

possibly as a hypothesis, but not from encoding absence alone.

Remark 1.1 (Current status). The results proved here are classification and non-identifiability results. No selected MTT action, covariant dark-sector stress tensor, cosmological solution, likelihood, or held-out numerical prediction is supplied. Thus this paper is interpretive and diagnostic. It is not a replacement for Λ CDM, particle dark matter, or a specified modified-gravity theory.

The central correction to version 1 is the distinction between three claims:

- (i) an encoding is unavailable in a reduced description;
- (ii) a covariant source or modified field equation is selected;
- (iii) that dynamics reproduces the observed dark-sector data.

Claim (i) does not imply (ii), and claim (ii) does not imply (iii).

1.1 Position in the MTT program

Program C distinguishes spacetime coordinates, internal bundles, connections, operators, spectral projectors, and reduced descriptions as different typed objects [4]. This paper adopts that discipline. An “encoding” is not a substance, a coordinate, a stress tensor, or an interaction merely because the same verbal label occurs in several descriptions.

The D-layer is therefore downstream of the following missing construction:

$$\text{MTT source data} \longrightarrow \text{selected covariant dynamics} \longrightarrow \text{cosmological observables.}$$

The first arrow is not supplied by naming an E8 or E9 regime.

2 A Typed Encoding-Availability Hypothesis

2.1 Spacetime and availability data

Let (Y_4, g) be a time-oriented Lorentzian spacetime. Let

$$\mathcal{E} = \{E_1, \dots, E_9\}$$

denote a fixed catalogue of reduced encoding types. Nothing in this notation identifies an E_i with a physical field or source.

Definition 2.1 (Availability profile). An *availability profile* is a map

$$\alpha : Y_4 \longrightarrow \mathcal{A},$$

where $\mathcal{A}$ records which encoding contracts are satisfied in each region. In the simplest bookkeeping model, $\mathcal{A} \subseteq \{0, 1\}^9$; a continuous profile may instead take values in $[0, 1]^9$ once a quantitative availability functional has been defined.

Definition 2.2 (Available encoding). An encoding E_i is available on an open set $U \subseteq Y_4$ if its declared domain, codomain, overlap, locality, and stability conditions are all satisfied on U . It is unavailable if at least one declared condition fails.

Remark 2.3. Availability is relative to an explicit encoding contract. It is not an observer-independent physical scalar until a law defining α from pre-projection data has been supplied.

2.2 The E8 and E9 labels

For the present hypothesis, E8 and E9 have only the following meanings.

Definition 2.4 (Relational profile, E8). An E8 profile retains a declared set of relations or transition data while some specified local descriptions are unavailable.

Definition 2.5 (Statistical profile, E9). An E9 profile retains a probability measure, state, or ensemble functional on a declared measurable state space while some specified local descriptions are unavailable.

The measure or state in the E9 definition is mandatory. Calling a description “statistical” without specifying its measurable space and probability law does not define an encoding.

Proposition 2.6 (Availability is not ontology). *Neither $\alpha_i = 0$ nor $\alpha_i = 1$ determines the existence or absence of a particle, field, fluid, stress tensor, or metric degree of freedom.*

Proof. The codomain of α records satisfaction of descriptive contracts. Particle content, fields, fluids, and metric dynamics belong to different typed spaces. No map from $\mathcal{A}$ to any of those spaces has been given, so no such conclusion follows. $\square$

3 The Availability-Only Non-Selection Theorem

The main mathematical result of this revision is that the old physical identifications were underdetermined.

Theorem 3.1 (Availability-only non-selection). *Let (Y_4, g) and an availability profile α be fixed. If the premises place no condition on an action, field equation, or source map, then they do not select a unique gravitational source or spacetime dynamics.*

Proof. Keep the same (Y_4, α) . One admissible completion may assign no additional action,

$$S_{\text{enc}}^{(0)}[g, \alpha] = 0, \quad T_{\mu\nu}^{(0)} = 0.$$

A second completion may assign

$$S_{\text{enc}}^{(\Lambda)}[g, \alpha] = - \int_{Y_4} \sqrt{-g} \Lambda_{\text{eff}} d^4x, \quad T_{\mu\nu}^{(\Lambda)} = -\Lambda_{\text{eff}} g_{\mu\nu},$$

where

$$T_{\mu\nu} := - \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{enc}}}{\delta g^{\mu\nu}}.$$

Both completions have the same availability profile, yet their sources and metric equations differ whenever $\Lambda_{\text{eff}} \neq 0$. Therefore availability data alone do not select the source or dynamics. $\square$

Corollary 3.2 (No source from absence). *The unavailability of local encodings does not by itself source curvature.*

Corollary 3.3 (No dark-sector identification from labels). *An E9 label alone does not derive dark matter, and an E8 label alone does not derive dark energy.*

3.1 Why E9 does not imply cold dark matter

A statistical state can describe many inequivalent effective media. At the level of a homogeneous stress tensor,

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu} + \pi_{\mu\nu},$$

the E9 label fixes none of ρ , p , u^μ , or $\pi_{\mu\nu}$.

Proposition 3.4 (Equation-of-state underdetermination). *The existence of an E9 statistical encoding does not imply*

$$p = 0, \quad c_s^2 = 0, \quad \pi_{\mu\nu} = 0,$$

and hence does not imply cold, pressureless, collisionless behavior.

Proof. Probability measures exist for dust, radiation, warm fluids, interacting fluids, and systems with anisotropic stress. These descriptions share the property “statistical” but have different equations of state, sound speeds, collision terms, and perturbation evolution. The label therefore cannot select one of them. $\square$

3.2 Why E8 does not imply acceleration

For a spatially homogeneous and isotropic solution of Einstein gravity,

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}.$$

Proposition 3.5 (Acceleration is dynamical). *An E8 relational encoding does not determine the sign of $\ddot{a}$.*

Proof. The sign depends on the source combination $\rho + 3p$, a cosmological term, or the corresponding terms in a modified metric equation. The definition of E8 specifies none of them. The same E8 profile is therefore compatible with accelerating, coasting, or decelerating solutions. $\square$

4 The Covariant Completion Contract

A physical promotion requires one of two typed routes.

4.1 Route A: an effective source action

Specify fields or source variables χ . If the availability profile is dynamical, include it among those variables and specify its equation of motion. A diffeomorphism-invariant functional then has the form

$$S_{\text{enc}}[g, \chi, \alpha].$$

and equations obtained by varying all dynamical arguments. Define

$$T_{\mu\nu}^{\text{enc}} := -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{enc}}}{\delta g^{\mu\nu}}.$$

The metric equation may then take the form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^{\text{enc}} \right).$$

Theorem 4.1 (On-shell covariant conservation). *If S_{enc} is diffeomorphism invariant and the Euler–Lagrange equations hold for every spacetime-dependent nonmetric argument, including α when it is dynamical, then*

$$\nabla^\mu T_{\mu\nu}^{\text{enc}} = 0.$$

Proof. Under an infinitesimal diffeomorphism generated by ξ^μ , the metric variation is $\delta g^{\mu\nu} = -2\nabla^{(\mu}\xi^{\nu)}$. The Euler–Lagrange terms of all nonmetric variables vanish on shell. Diffeomorphism invariance and integration by parts then give

$$0 = \delta S_{\text{enc}} = - \int_{Y_4} \sqrt{-g} (\nabla^\mu T_{\mu\nu}^{\text{enc}}) \xi^\nu d^4x.$$

Since ξ^ν is arbitrary, the result follows. $\square$

Remark 4.2 (External availability profiles). If α is prescribed rather than varied, its Euler–Lagrange response does not vanish in the Noether identity. Separate conservation of $T_{\mu\nu}^{\text{enc}}$ then need not hold. A completed model must either make α dynamical, prove that its response term vanishes on the claimed background, or state the resulting covariant exchange law.

4.2 Route B: a modified metric equation

Alternatively, specify a covariant tensor

$$\mathcal{E}_{\mu\nu}[g, \alpha, \chi] = 8\pi G T_{\mu\nu}^{\text{visible}}.$$

This route must state its initial-value problem, propagating degrees of freedom, stability conditions, and divergence identity. In particular, consistency requires

$$\nabla^\mu \mathcal{E}_{\mu\nu} = 0$$

on the declared solution space, or an explicitly balanced exchange equation.

Remark 4.3. Calling the proposal “unchanged gravity” commits it to Route A with a nonzero effective source. Calling it “not a new source” generally commits it to Route B. Version 1 implicitly used both descriptions. A completed model must choose and execute one.

4.3 Selection and parameter provenance

Whichever route is used, an MTT derivation must provide:

1. the domain and codomain of the source map;
2. the pre-projection data on which it depends;
3. the selector that fixes its functional form;
4. all dimensional scales and dimensionless parameters;
5. the conservation or exchange identity;
6. a well-posed and stable evolution problem;
7. an observable map and uncertainty model;
8. a separation of fitted, calibrated, and held-out quantities.

A freely chosen action that reproduces cosmology would be a legitimate effective model, but not yet an MTT selection theorem.

5 Dark-Matter Promotion Gates

The corrected hypothesis is:

Conjecture 5.1 (E9 dark-matter hypothesis). There exists a selected MTT source construction whose relevant reduced description factors through an E9 statistical state and whose effective dynamics pass the dark-matter gates below.

This conjecture does not say that dark matter *is* missing description. It proposes a possible factorization of a future source law.

5.1 Background and perturbation requirements

At minimum, a candidate must derive a background density and pressure and the linear perturbation variables

$$\delta\rho, \quad \delta p, \quad \theta, \quad \pi_{\mu\nu}.$$

To reproduce a cold-dark-matter limit over a stated range, it must show, with error bounds,

$$|w| = \left| \frac{p}{\rho} \right| \ll 1, \quad c_s^2 = \frac{\delta p}{\delta \rho} \ll 1,$$

and control anisotropic stress and any collision or drag term. These properties cannot be inserted by redefining “statistical.”

5.2 Independent observational gates

The candidate must be executed against:

- DM1. CMB:** acoustic-peak heights, lensing, and the transfer functions through recombination;
- DM2. BAO and growth:** background distances and growth observables such as $f\sigma_8$;
- DM3. weak and strong lensing:** the relation between dynamical and lensing potentials;
- DM4. cluster collisions:** spatial offsets among lensing mass, galaxies, and baryonic gas, including Bullet-Cluster-type systems [1];
- DM5. nonlinear structure:** halo abundance, profiles, substructure, and the matter power spectrum;
- DM6. local tests:** any transition scale at which the proposed availability law changes behavior.

Planck’s CMB analysis constrains a successful baseline model far beyond a generic statement of gravitational clustering [5]. DES weak-lensing and galaxy-clustering analyses add independent structure-growth information [2]. Passing only rotation-curve or direct-detection considerations is therefore insufficient.

6 Dark-Energy Promotion Gates

The corrected second hypothesis is:

Conjecture 6.1 (E8 dark-energy hypothesis). There exists a selected MTT source or metric construction whose relevant reduced description factors through E8 relational data and whose effective dynamics pass the dark-energy gates below.

6.1 Background requirement

Under Route A in Einstein gravity, accelerated expansion requires an effective source satisfying

$$\rho_{\text{eff}} + 3p_{\text{eff}} < 0$$

over the relevant epoch. Under Route B, the modified metric terms must imply the corresponding acceleration while preserving consistency and stability. The function $H(z)$ or an equivalent background solution must be derived.

6.2 Perturbations and stability

The model must also specify dark-sector perturbations or prove their absence. It must control:

$$\delta\rho_{\text{eff}}, \quad \delta p_{\text{eff}}, \quad c_s^2, \quad \pi_{\mu\nu},$$

as applicable, and exclude ghost, gradient, and unacceptable strong-coupling instabilities on its claimed domain.

6.3 Independent observational gates

A physical model must confront:

- DE1.** supernova luminosity distances;
- DE2.** BAO distance ratios and expansion history;
- DE3.** CMB distance, lensing, and early-time consistency;
- DE4.** structure growth and weak lensing;
- DE5.** the equation-of-state history $w(z)$ or the model’s direct alternative;
- DE6.** correlations among all fitted probes and a held-out test.

The observational target is not frozen to one verbal description. Planck found broad success for base Λ CDM in its final CMB analysis [5], while DESI DR2 reports increasingly precise BAO constraints and examines departures from a cosmological constant in combination with other probes [3]. A future MTT completion must be tested against the data likelihoods, not declared compatible from the E8 label.

7 What the Hypothesis Does and Does Not Explain

7.1 Logical economy

The availability viewpoint establishes a logical possibility: gravitationally inferred phenomena need not, before a source theory is known, be identified with one particular microscopic ontology. This is a useful warning against premature identification. It does not show that no new particles, fields, or modified dynamics are required by nature.

Proposition 7.1 (Non-ontology is not an explanation). *Showing that a new microscopic object is not logically forced is weaker than constructing a model that reproduces the observations.*

Proof. The former excludes a logical implication. The latter requires a selected dynamics and agreement with data. Theorem 3.1 shows that the availability premise leaves those dynamics undetermined. $\square$

7.2 Direct detection

The absence of a confirmed non-gravitational dark-matter signal does not by itself favor the E9 hypothesis. Many particle models predict suppressed or experiment-dependent signals, and the E9 proposal has no detection-rate calculation. Conversely, a robust non-gravitational signal would rule out only those completed E9 models that explicitly predict its absence.

7.3 Modified gravity

Before Route A or Route B is specified, the proposal cannot be distinguished from an effective source theory or a modified-gravity theory. That distinction is made by equations, not by interpretation.

7.4 Homogeneity and clustering

Relational data need not be homogeneous, and statistical data need not cluster. Homogeneity, isotropy, and clustering are properties of a state and its dynamics. They are not consequences of the words “relational” and “statistical.”

8 Relation to Current MTT Results

8.1 Typed realization results

The revised Program C supplies a typed language for internal geometry, bundles, connections, operators, projectors, and reduced descriptions [4]. D1 uses that language negatively: no internal operator or availability label is promoted to a spacetime source without a declared map.

8.2 Results that do not close D1

The following results, even when valid in their own scopes, do not supply the missing cosmological source:

- existence or stability of a fixed point;
- a finite internal branch selector;
- a shared line bundle or central phase circle;
- an internal rank profile or finite matrix;
- a coherent spectral projector;
- a local world-in-world component decomposition.

To contribute to D1, such data must enter a selected action or field equation and survive dimensional reduction with a proved source and observable map.

8.3 Current readiness tier

We use the following audit scale:

Tier	Requirement
DS0	Typed availability hypothesis and non-identifiability theorem.
DS1	Selected covariant source map or modified metric equation.
DS2	Stable homogeneous cosmological solutions.
DS3	Linear perturbations and observable transfer functions.
DS4	Nonlinear lensing, structure, and cluster-collision execution.
DS5	Joint likelihood, provenance, uncertainty, and held-out tests.

Remark 8.1 (Audited readiness). This paper closes DS0. Tiers DS1–DS5 remain open. No numerical dark-sector result is claimed.

9 A Non-Circular Research Program

The most efficient continuation is ordered by logical dependence.

9.1 Step 1: select the source law

Construct a pre-projection functional whose reduction produces either S_{enc} or $\mathcal{E}_{\mu\nu}$. Prove covariance, specify every parameter, and show why the selected form is preferred over the counterexamples in Theorem 3.1.

9.2 Step 2: derive conservation and well-posedness

Use Theorem 4.1 or the corresponding modified-gravity identity. Establish the initial-value problem and stability domain before fitting cosmology.

9.3 Step 3: solve the background

Derive $H(z)$, the density evolution, and any transition law for availability. Do not infer these functions from the data and then call the construction a prediction.

9.4 Step 4: execute perturbations

Derive scalar, vector, and tensor perturbations as applicable. Compute the CMB, lensing, BAO, and growth observables using a reproducible solver.

9.5 Step 5: execute nonlinear and held-out tests

Confront cluster collisions and nonlinear structure, publish the likelihood and covariance treatment, and reserve at least one informative data product as a held-out test.

10 Falsifiability

The uncompleted availability hypothesis is not yet a quantitative physical model and therefore has no unique observational prediction. A completed model becomes falsifiable through its selected equations.

Definition 10.1 (Model-level falsifier). A model-level falsifier is a predeclared observable or consistency condition whose failure lies outside the model’s propagated uncertainty and declared validity domain.

Examples include:

- failure of covariant conservation or the declared exchange law;
- a ghost or gradient instability in the claimed cosmological epoch;
- incompatible CMB peak or lensing spectra;
- incorrect BAO or supernova distances;
- incorrect growth or weak-lensing predictions;
- failure in merging clusters or nonlinear structure;
- a held-out likelihood decisively worse than the stated comparator.

The absence of a direct-detection signal is not, by itself, a positive prediction of the present paper.

11 Version 2 Revision Record

Version 2 is a structural correction, not a boundary-only erratum.

1. The title and abstract now label the proposal as a projection-first dark-sector hypothesis.
2. The E8 and E9 profiles are typed as reduced-description data.
3. The availability-only non-selection theorem proves that these profiles do not determine a stress tensor or metric dynamics.
4. The former claims that E9 explains pressureless clustering and non-interaction are withdrawn.
5. The former claims that E8 produces homogeneity and acceleration are withdrawn.
6. A covariant action/source route and a modified-metric route are stated separately.
7. On-shell conservation is proved for a diffeomorphism-invariant source action.
8. CMB, BAO, lensing, cluster-collision, structure-growth, and equation-of-state tests are made mandatory before physical promotion.
9. Compatibility with precision cosmology is no longer asserted without execution.
10. The current result is recorded as tier DS0; all physical tiers DS1–DS5 remain open.

12 Conclusion

The projection-first idea survives in a precise form. MTT may classify regimes in which different reduced encodings are available, and E8/E9 labels may help organize future source constructions. What does not survive is the claim that missing encodings already explain the observed dark sector.

Availability does not determine stress-energy. Statistical description does not determine cold dark matter. Relational description does not determine accelerated expansion. The next decisive object is a selected covariant source action or modified field equation, followed by background, perturbative, nonlinear, and held-out observational execution. Until that chain is supplied, Program D1 is a disciplined hypothesis and research contract rather than a dark-sector theory.

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